Week 3 EDA + Data Strategy Finalisation

1. Class Imbalance

The following table reports imbalance at volume, slice and pixel levels.

class	volume_count	slice_count	pixel_count	pixel_fraction
1	23	181	301856	0.0560379
2	127	888	2.36379e+06	0.438825
3	104	713	709872	0.131784
4	109	976	787323	0.146162
5	70	765	1.2238e+06	0.227192

2. Low-Volume Classes

The following classes contain fewer than 50 training volumes and should be considered high-risk for unstable segmentation performance.

class	volume_count	slice_count	pixel_count	pixel_fraction	high_risk_low_dice
1	23	181	301856	0.0560379	True

3. Dataset Split

split	volume_count
train	134
validation	28
active_learning	30

4. Zero Overlap Verification

```
ZERO OVERLAP CHECK
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Train ∩ Validation: 0
Train ∩ AL: 0
Validation ∩ AL: 0

ZERO OVERLAP: True
```

5. Data Strategy

The analysis distinguishes three levels of class imbalance:

1. Volume-level imbalance
2. Slice-level imbalance
3. Pixel-level imbalance

These measurements are treated separately because they affect different modelling decisions.

Volume-level imbalance affects whether the model has sufficient independent patient/volume examples.

Slice-level imbalance affects the number of training examples containing each class.

Pixel-level imbalance reflects the spatial dominance of large background or hemorrhage regions.

Training, validation and active-learning pools are split at volume level to prevent slices from the same volume appearing in different partitions.